AWS VPC Peering: Student Project Report

Oct 21, 2026

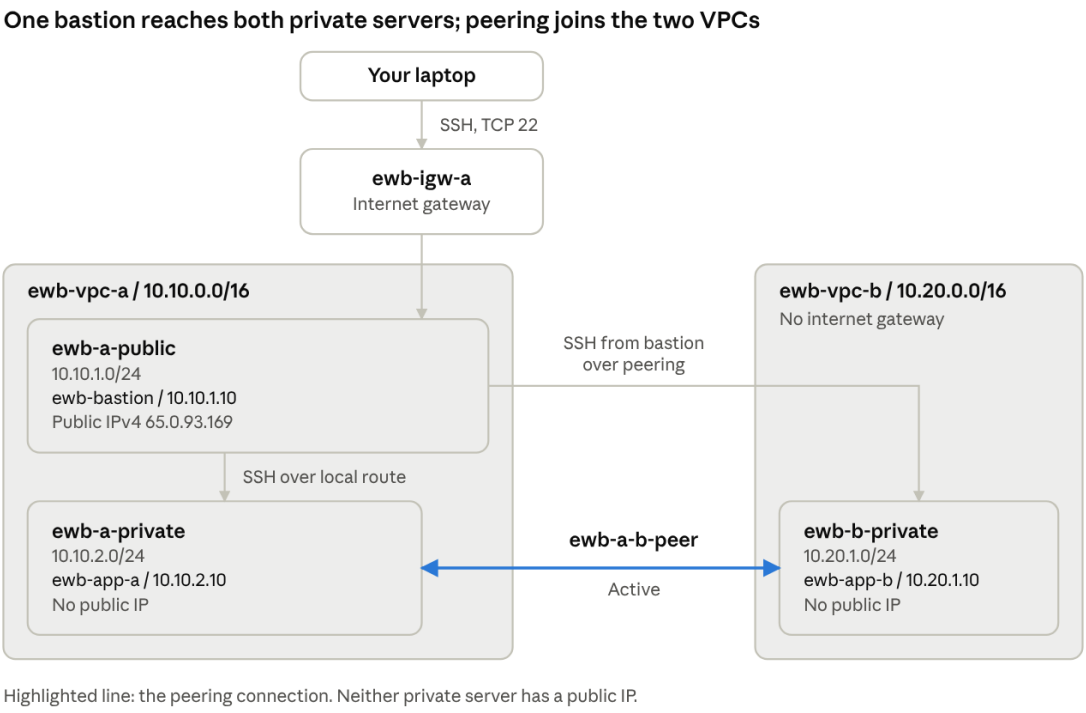
Two private VPCs connected with VPC peering, with a bastion host for administration and test evidence of private connectivity.

Project details

Field	Value
Student name	Jiya Mary Jacob
Batch	AWS Internship Batch 1
Project title	AWS VPC Peering
Region	Asia Pacific (Mumbai) / ap-south-1
Availability Zone	ap-south-1a
Start date	28 September 2026
Completion date	21 October 2026
Tags on every resource	Project = EWB-VPC-Peering, Student = Jiya

Architecture

One public bastion in VPC A reaches two private servers, and VPC A and VPC B are joined by one peering connection.



architecture · 2 VPCs, 3 instances, 1 peering connection

Administration path: laptop to bastion to App-A or App-B using SSH ProxyJump. Project traffic: App-A and App-B over private IPs through the peering connection. VPC B has no internet gateway.

Resource inventory

All resources are tagged Project = EWB-VPC-Peering and Student = Jiya. VPC B has no internet gateway.

VPCs, subnets and internet gateway

Resource (Name tag)	ID	CIDR / detail
ewb-vpc-a	vpc-074f44afffd33e8b0	10.10.0.0/16
ewb-vpc-b	vpc-071028f353ae50f21	10.20.0.0/16
ewb-a-public	subnet-00a5804e737fb0429	10.10.1.0/24, ap-south-1a
ewb-a-private	subnet-0234c35664492b902	10.10.2.0/24, ap-south-1a
ewb-b-private	subnet-035b918b997acd8f7	10.20.1.0/24, ap-south-1a
ewb-igw-a	igw-07c4520711c1268fa	Attached to ewb-vpc-a

Instances

All three run Amazon Linux 2023 on t3.micro.

Name	Instance ID	VPC / subnet	Private IP	Public IPv4	Security group
ewb-bastion	i-007009e53589bd038	ewb-vpc-a / ewb-a-public	10.10.1.10	65.0.93.169	ewb-sg-bastion
ewb-app-a	i-06031e6f18aa79a9f	ewb-vpc-a / ewb-a-private	10.10.2.10	None	ewb-sg-app-a
ewb-app-b	i-0c07959a5eb16b3be	ewb-vpc-b / ewb-b-private	10.20.1.10	None	ewb-sg-app-b

Security groups and root volumes

Resource	Security group ID	Root volume ID
ewb-bastion	sg-012fdb869c6b1077f (ewb-sg-bastion)	vol-0afcdea04bad7d627
ewb-app-a	sg-0a28c745319e85766 (ewb-sg-app-a)	vol-03efac937c3c63e15
ewb-app-b	sg-002f9b9e3f8015ec8 (ewb-sg-app-b)	vol-0426b221ab7fd62dd

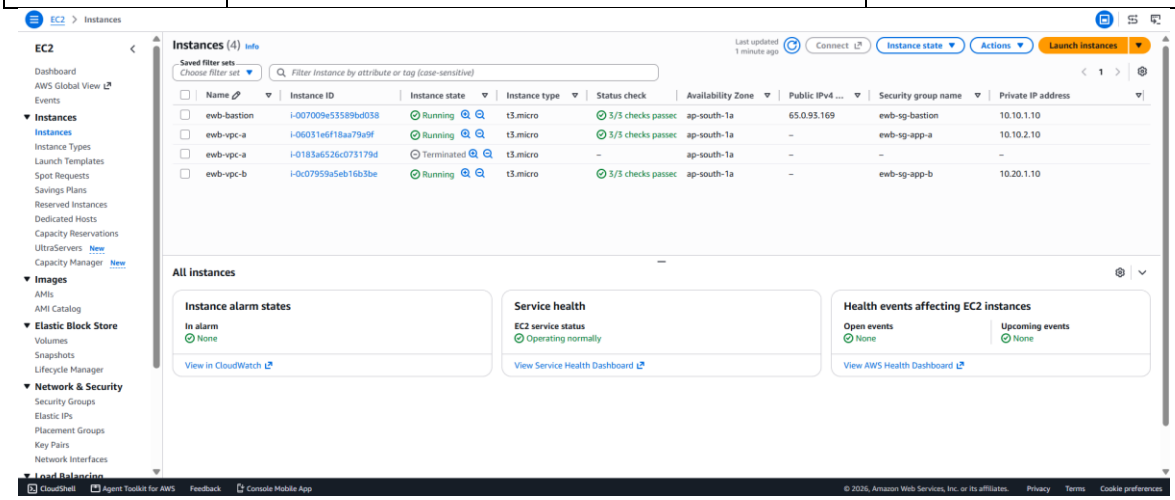


Figure 1. EC2 instance list: ewb-bastion, ewb-app-a and ewb-app-b running with 3/3 status checks passed, private IPs and the bastion public IPv4 (Checkpoint 02).

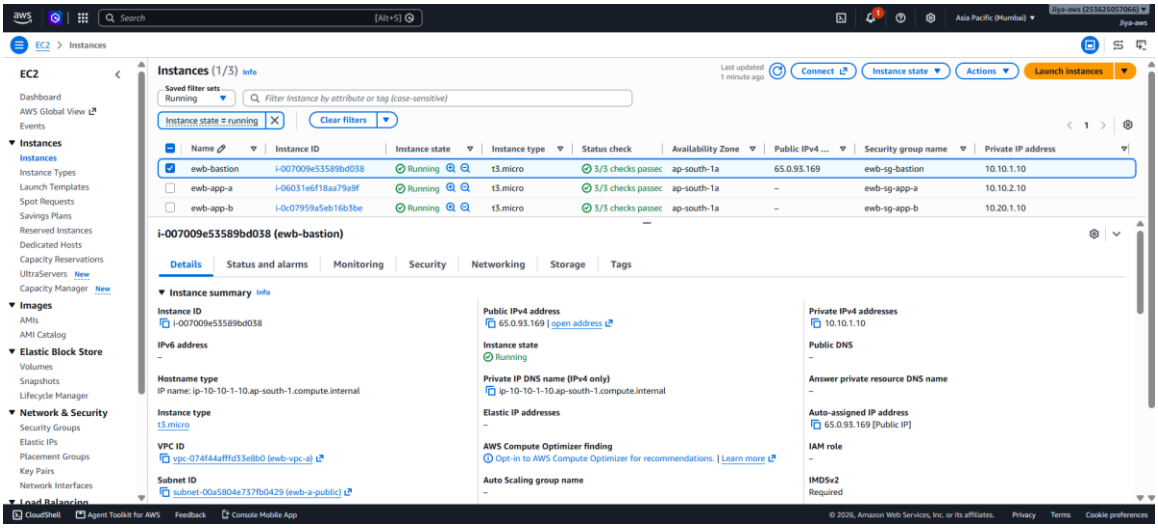


Figure 2. ewb-bastion instance details: private IP 10.10.1.10, public IPv4 65.0.93.169, in ewb-vpc-a / ewb-a-public.

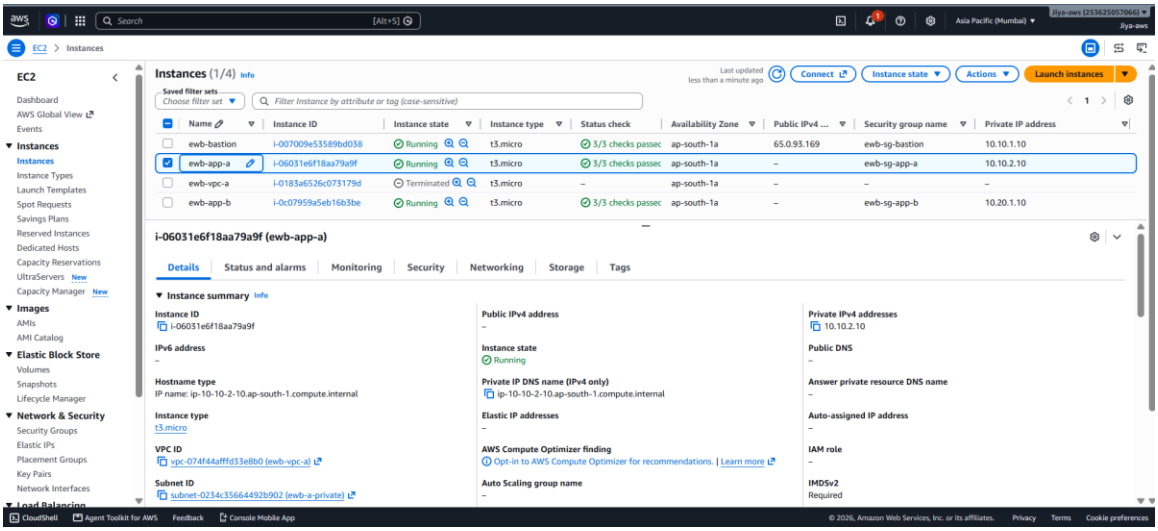


Figure 3. ewb-app-a instance details: private IP 10.10.2.10, in ewb-vpc-a / ewb-a-private.

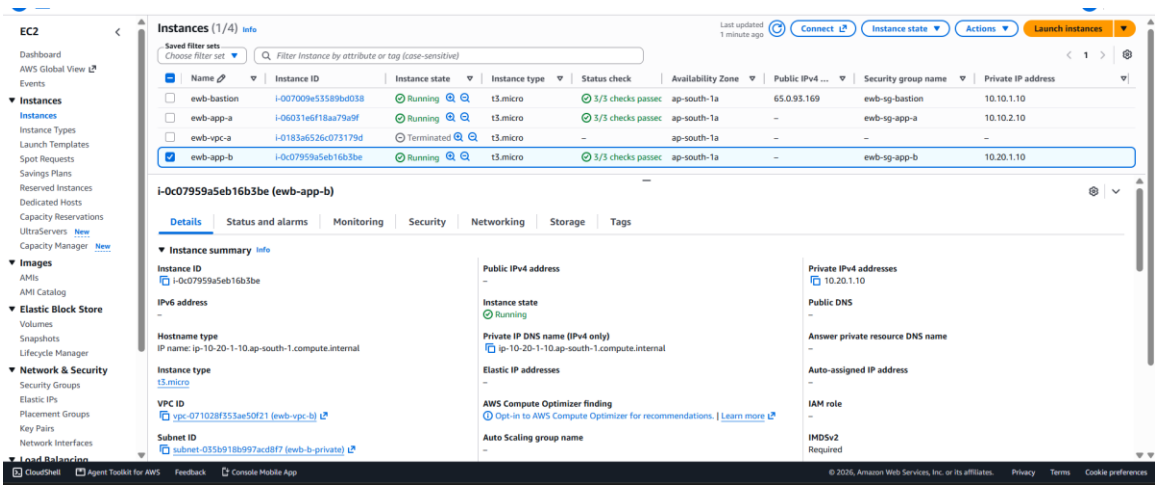


Figure 4. ewb-app-b instance details: private IP 10.20.1.10, in ewb-vpc-b / ewb-b-private.

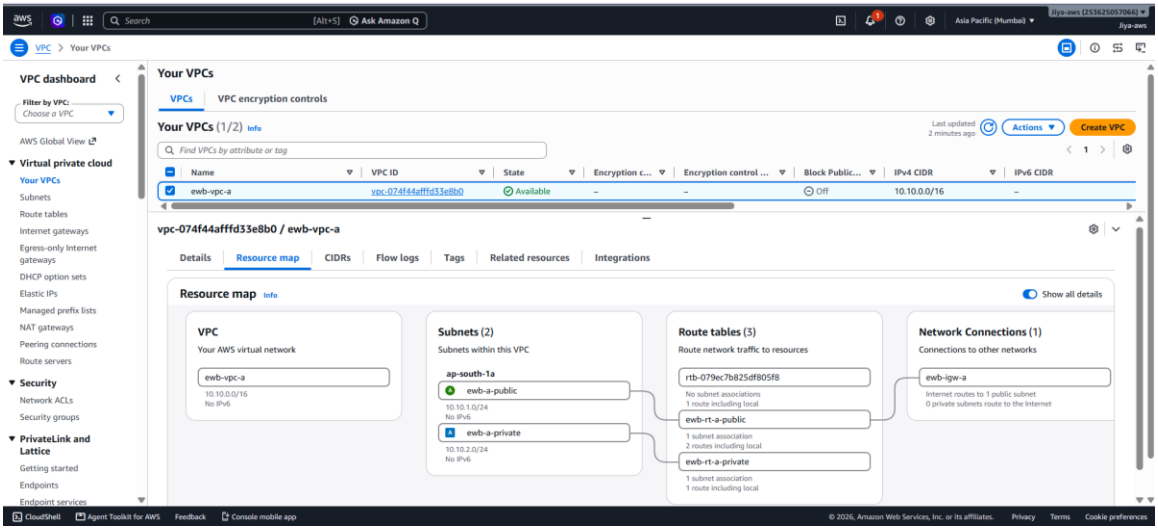


Figure 5. VPC resource map for ewb-vpc-a: two subnets, three route tables and the internet gateway.

Network build: route tables

Each subnet is explicitly associated with its own route table. Peering routes were added only after the peering connection became Active.

Route table	ID	Associated subnet	Routes
ewb-rt-a-public	rtb-0211d8a5965a3eda4	ewb-a-public	10.10.0.0/16 local; 0.0.0.0/0 igw-07c4520711c1268fa; 10.20.0.0/16 pcx-018895e855a398629
ewb-rt-a-private	rtb-0af2a2caab36a0585	ewb-a-private	10.10.0.0/16 local; 10.20.0.0/16 pcx-018895e855a398629
ewb-rt-b-private	rtb-04039a2202e5eb420	ewb-b-private	10.20.0.0/16 local; 10.10.0.0/16 pcx-018895e855a398629

The main route tables of both VPCs (rtb-079ec7b825df805f8 and rtb-0549dcd7b622b66b9) were left unchanged and have no explicit subnet associations.

The public subnet needs the 10.20.0.0/16 peering route because ProxyJump makes the bastion open its own connection to App-B. The more specific peering route is chosen over the internet default route.

ewb-rt-a-public

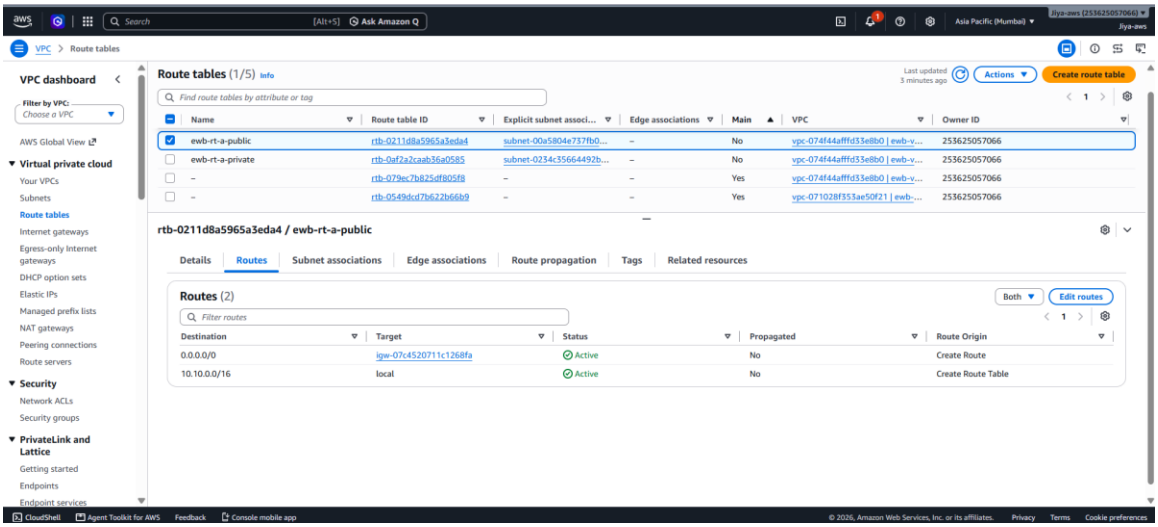


Figure 6. ewb-rt-a-public, Routes tab before peering: local route and 0.0.0.0/0 to the internet gateway.

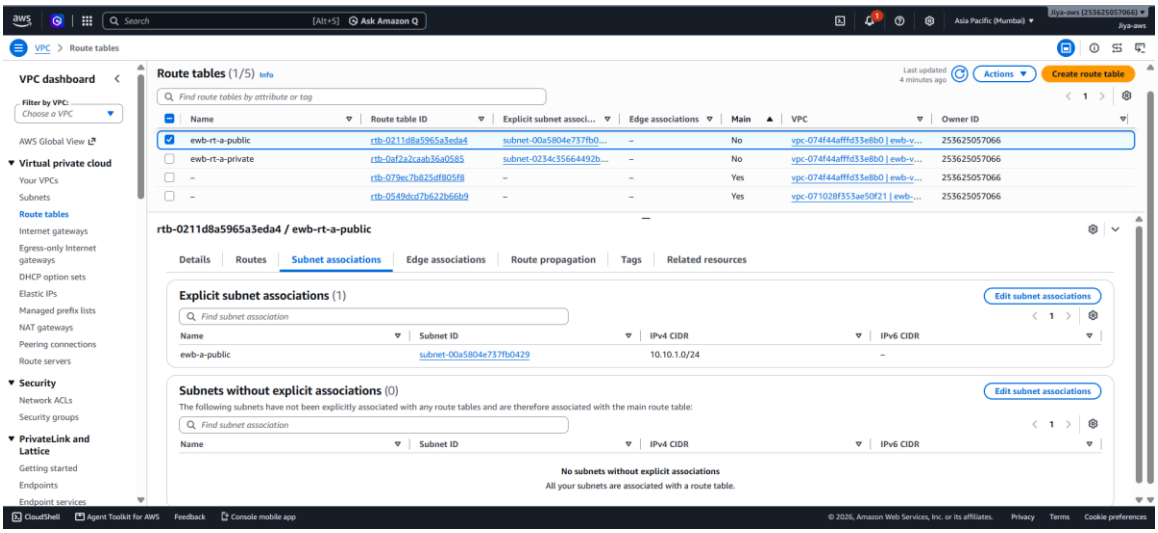


Figure 7. ewb-rt-a-public, Subnet associations tab before peering: ewb-a-public associated explicitly (Checkpoint 01).

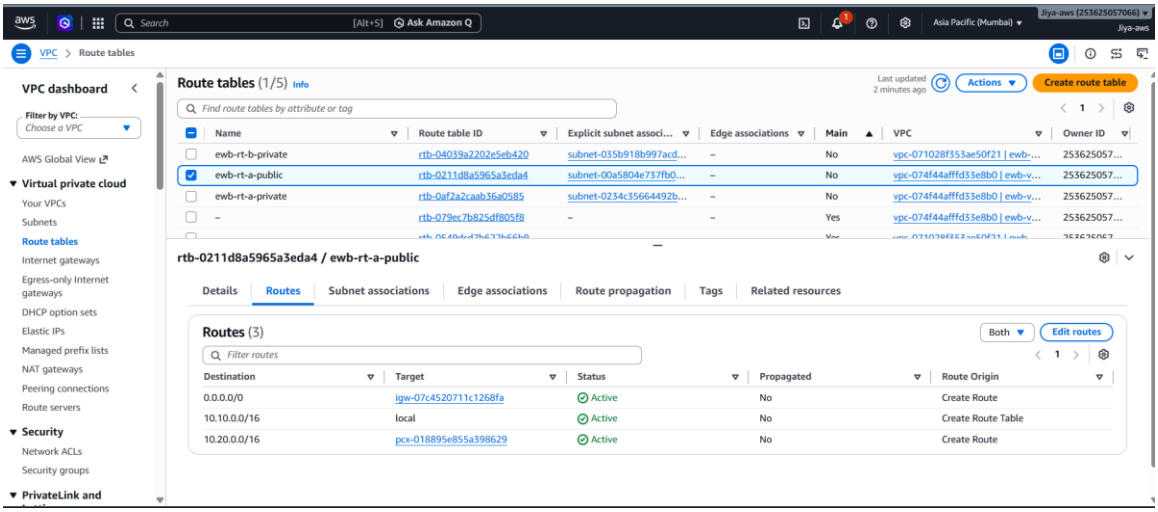


Figure 8. ewb-rt-a-public, Routes tab after peering: the 10.20.0.0/16 route to pcx-018895e855a398629 added (Checkpoint 04).

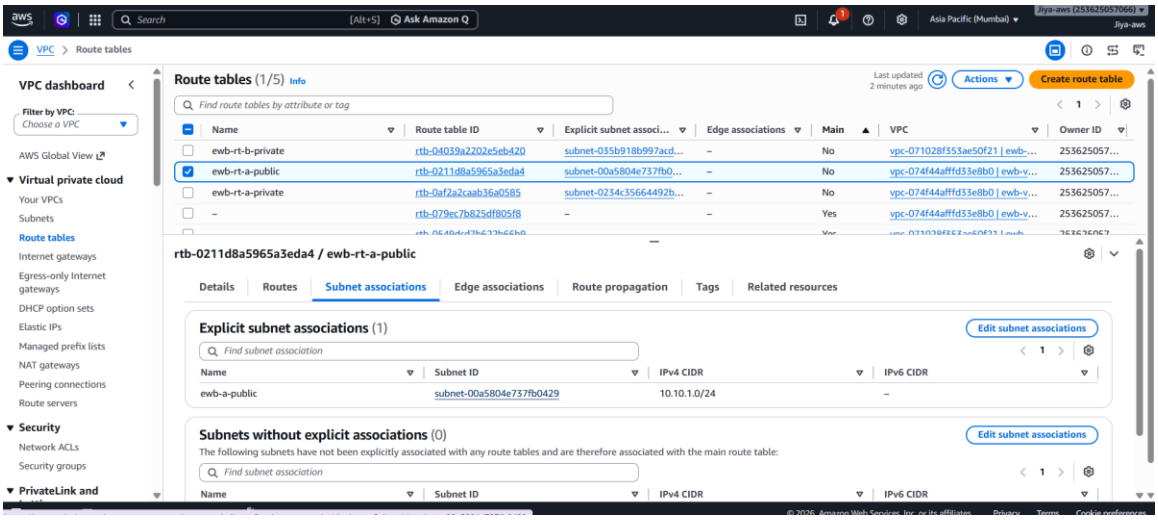


Figure 9. ewb-rt-a-public, Subnet associations tab after peering.

ewb-rt-a-private

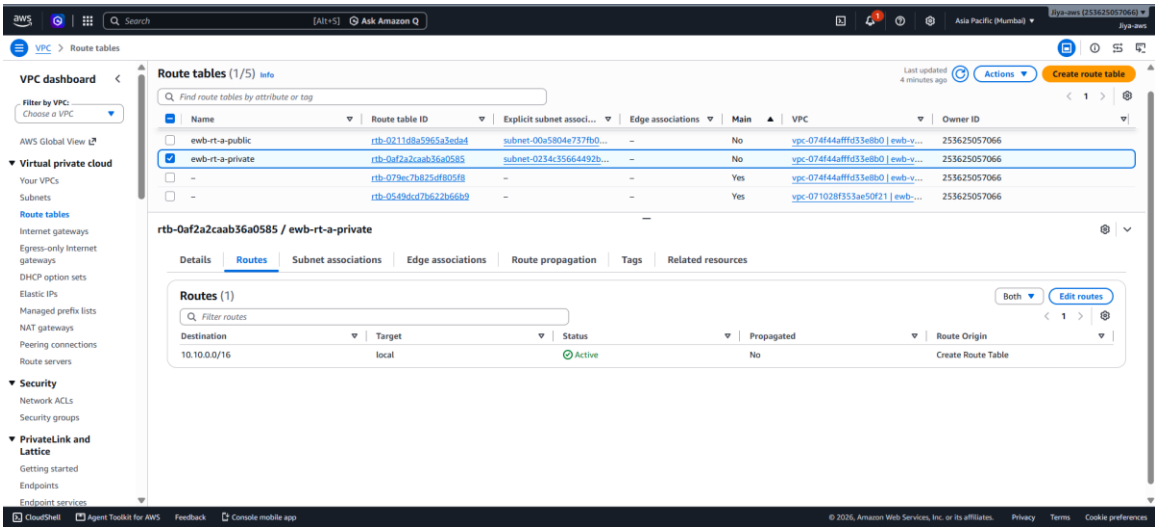


Figure 10. ewb-rt-a-private, Routes tab before peering: local route only (Checkpoint 01).

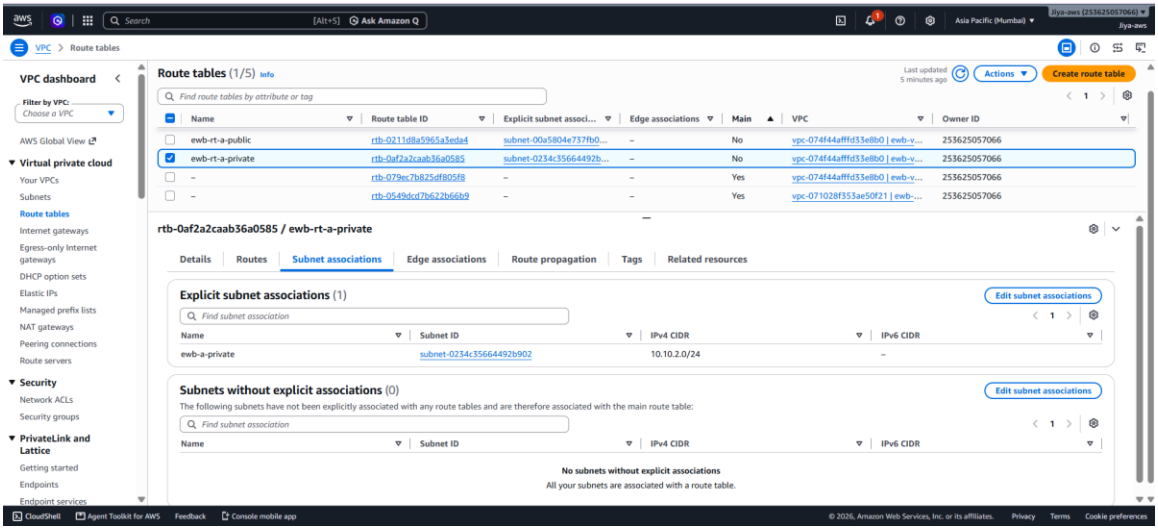


Figure 11. ewb-rt-a-private, Subnet associations tab before peering: ewb-a-private associated explicitly (Checkpoint 01).

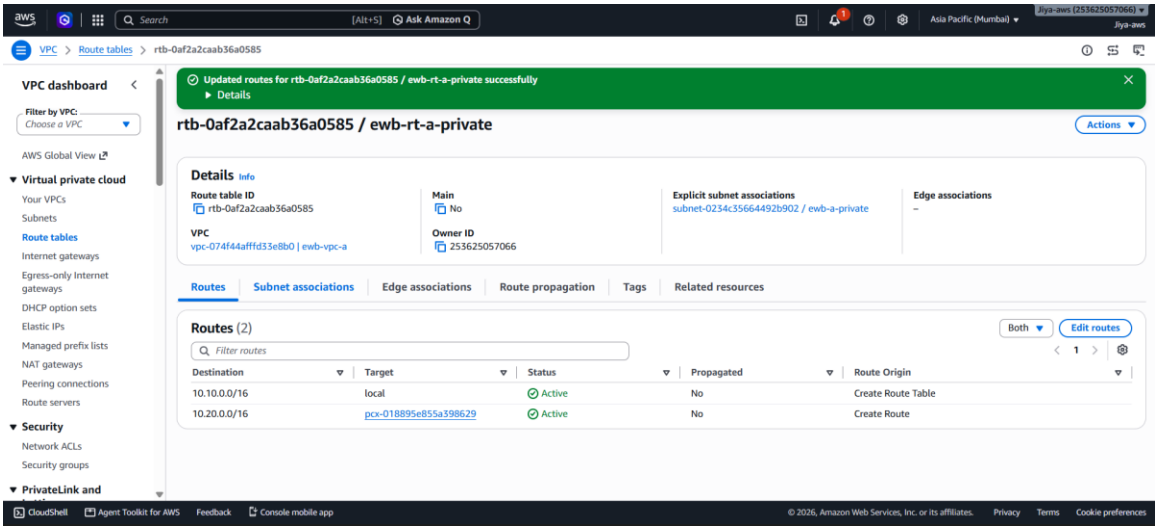


Figure 12. ewb-rt-a-private, Routes tab after peering: route update confirmed, 10.20.0.0/16 to pcx-018895e855a398629 (Checkpoint 04).

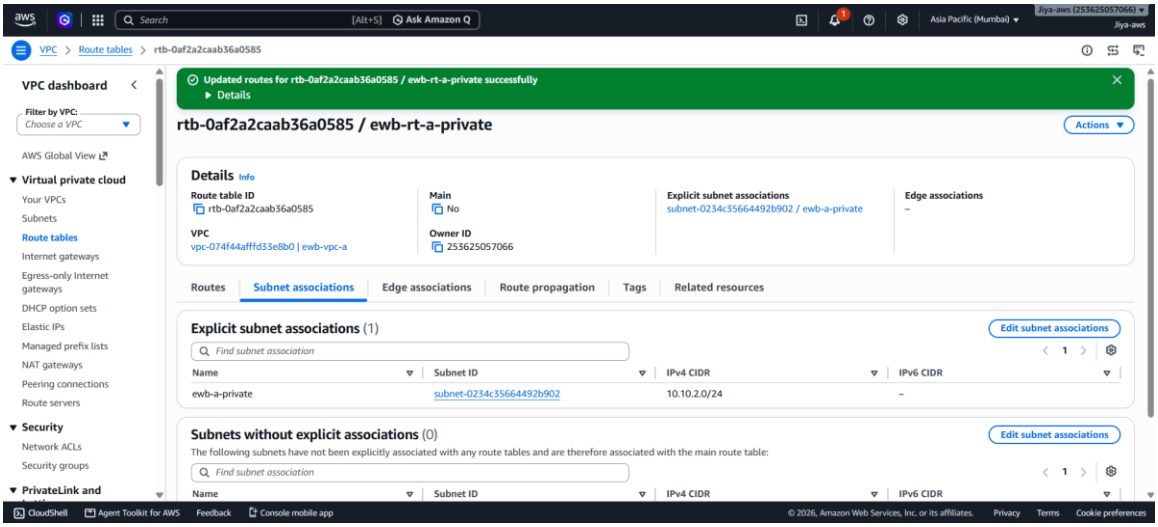


Figure 13. ewb-rt-a-private, Subnet associations tab after the route update.

ewb-rt-b-private

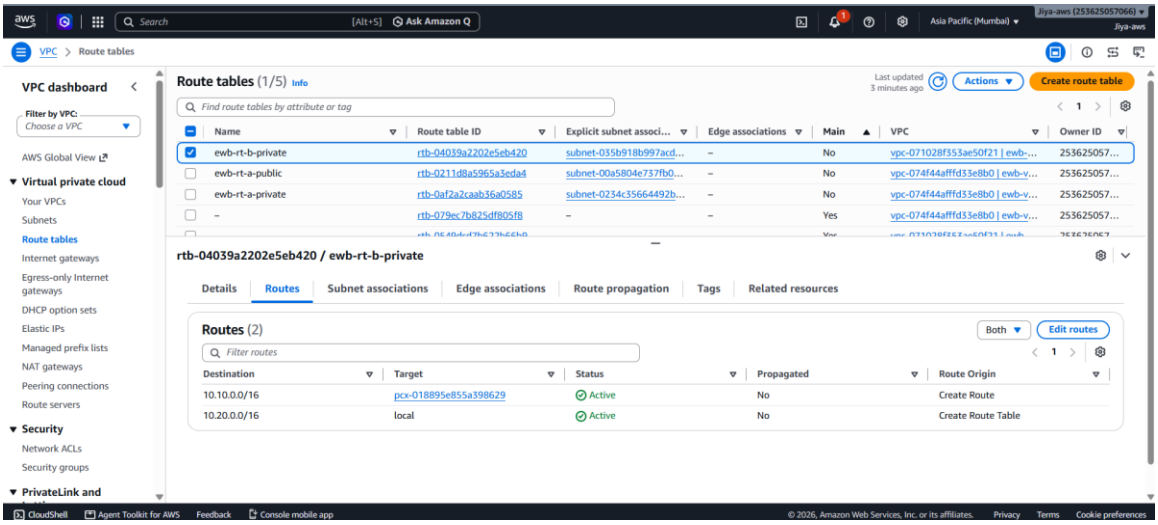


Figure 14. ewb-rt-b-private, Routes tab: local route and the return route 10.10.0.0/16 to pcx-018895e855a398629.

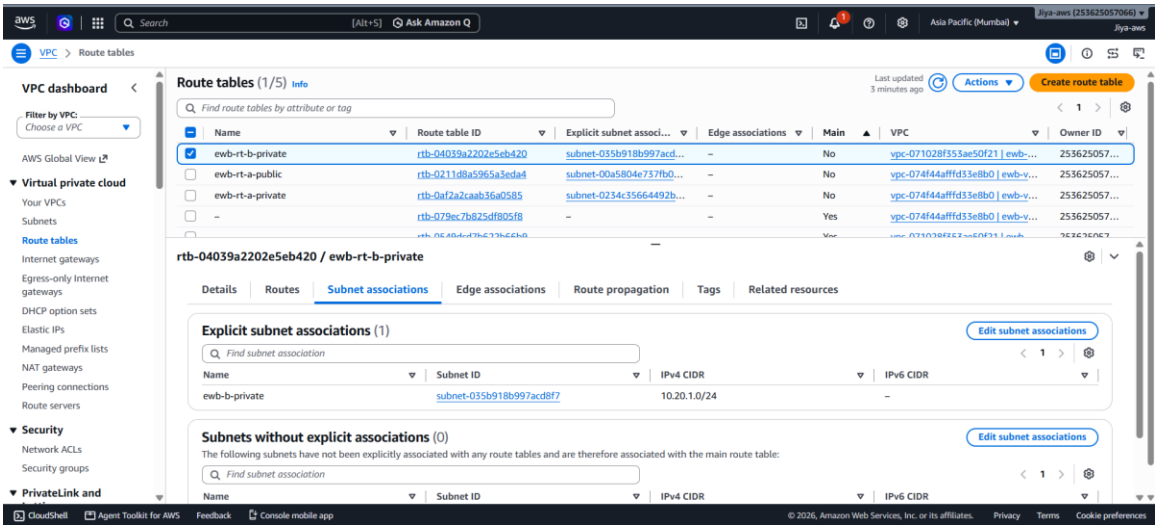


Figure 15. ewb-rt-b-private, Subnet associations tab: ewb-b-private associated explicitly.

Security groups

SSH and ping are allowed only from the specific lab hosts. Nothing is open to 0.0.0.0/0. Outbound rules keep the default allow-all.

Security group	Type	Port / protocol	Source
ewb-sg-bastion	SSH	TCP 22	65.0.93.169/32 (my public IP)
ewb-sg-app-a	SSH	TCP 22	10.10.1.10/32 (bastion private IP)
ewb-sg-app-a	Custom ICMP, IPv4	Echo request	10.20.1.10/32 (App-B)

Security group	Type	Port / protocol	Source
ewb-sg-app-b	SSH	TCP 22	10.10.1.10/32 (bastion private IP)
ewb-sg-app-b	Custom ICMP, IPv4	Echo request	10.10.2.10/32 (App-A)

VPC peering connection

The request was accepted manually, which is required even when both VPCs are in the same account.

Field	Value
Name	ewb-a-b-peer
Peering connection ID	pcx-018895e855a398629
Requester VPC	ewb-vpc-a / 10.10.0.0/16
Accepter VPC	ewb-vpc-b / 10.20.0.0/16
Account / Region	My account / This Region
Status	Active

Test evidence

All five tests passed. Hostnames recorded with `hostname` and `hostname -I`: App-A is ip-10-10-2-10.ap-south-1.compute.internal (10.10.2.10) and App-B is ip-10-20-1-10.ap-south-1.compute.internal (10.20.1.10).

Test	Host and command	Result
01 Baseline	App-A, before peering: `ping -c 4 10.20.1.10`	No replies, 100% packet loss
02 Forward path	App-A, after routes: `ping -c 4 10.20.1.10`	4 transmitted, 4 received, 0% packet loss
03 Reverse path	App-B, after routes: `ping -c 4 10.10.2.10`	4 transmitted, 4 received, 0% packet loss
04 Private SSH	Laptop: `ssh ewb-app-b`	Shell opened on App-B through the bastion
05 Isolation	EC2 details of App-A and App-B	Public IPv4 address empty on both

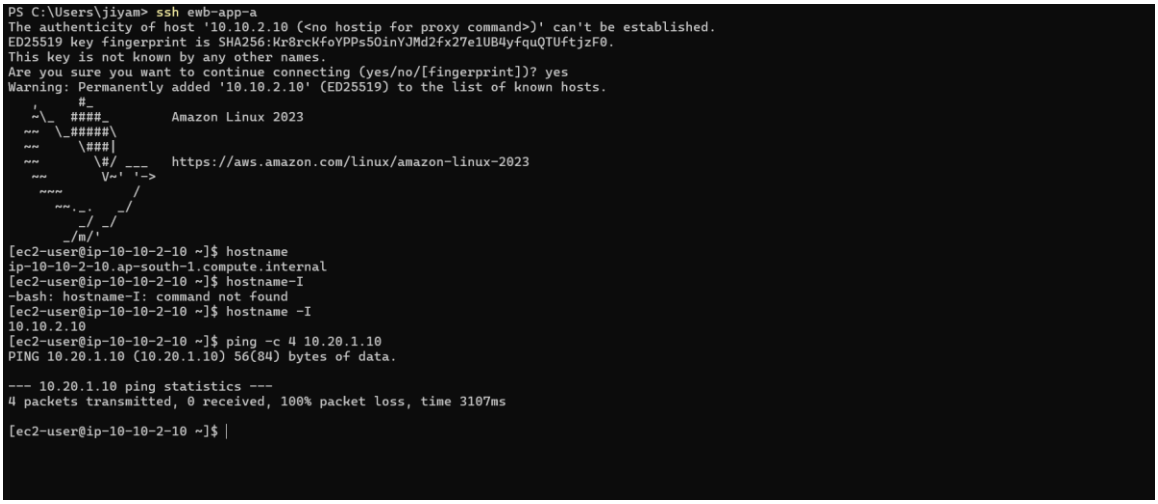


Figure 16. Test 01, App-A (ip-10-10-2-10), before peering: ping -c 4 10.20.1.10 gets 0 replies, 100% packet loss. hostname and hostname -I shown.

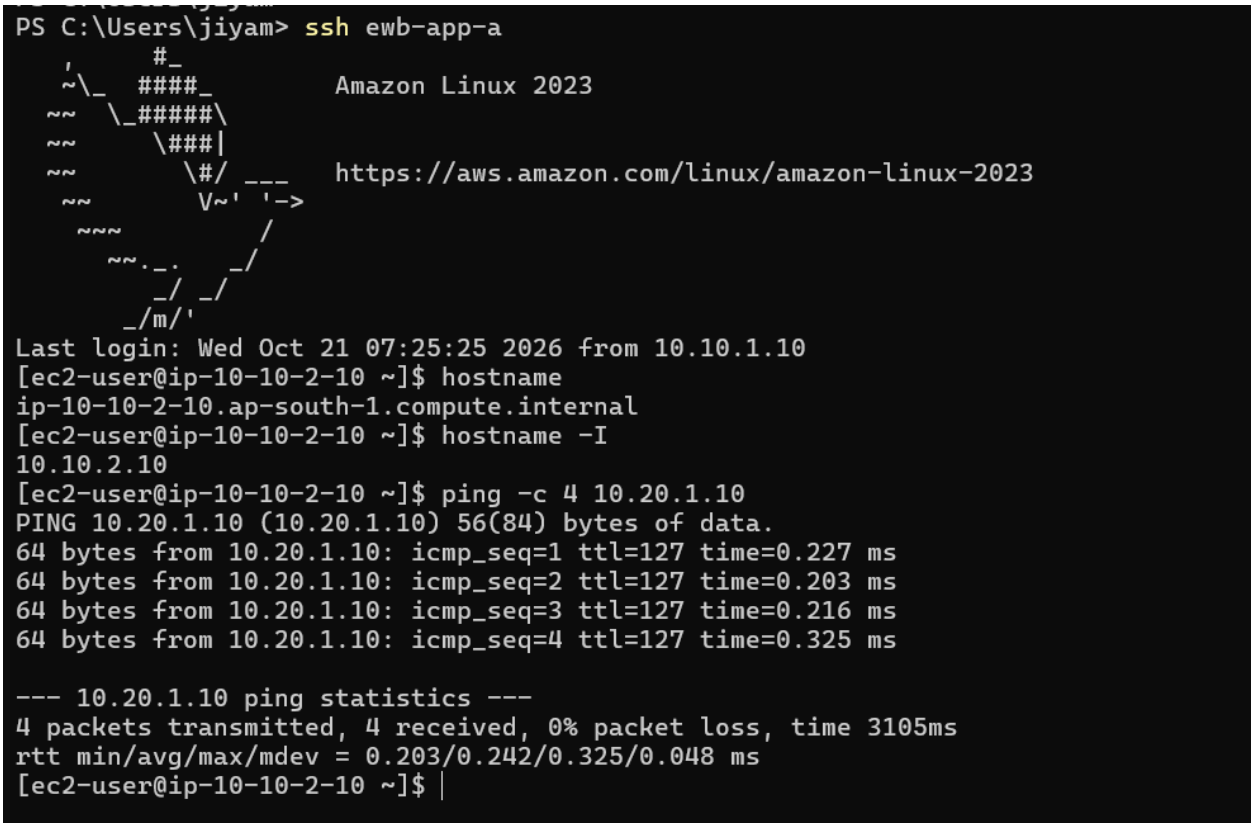


Figure 17. Test 02, App-A, after routes: ping -c 4 10.20.1.10 gets 4 of 4 replies, 0% packet loss.

```
The authenticity of host '10.20.1.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:2Nu+mf2s/KleZbTlN56HJ7K3NMrZls7MTS2N1ZxXMlw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-10 ~]$ hostname
ip-10-20-1-10.ap-south-1.compute.internal
[ec2-user@ip-10-20-1-10 ~]$ hostname -I
10.20.1.10
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.215 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.216 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.225 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.215 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3120ms
rtt min/avg/max/mdev = 0.215/0.217/0.225/0.004 ms
[ec2-user@ip-10-20-1-10 ~]$
```

Figure 18. Test 03, App-B (ip-10-20-1-10), after routes: ping -c 4 10.10.2.10 gets 4 of 4 replies, 0% packet loss. hostname and hostname -I shown.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\jiyam> ssh ewb-app-b
The authenticity of host '10.20.1.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:2Nu+mf2s/KleZbTlN56HJ7K3NMrZls7MTS2N1ZxXMlw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-10 ~]$
```

Figure 19. Test 04, laptop: ssh ewb-app-b opens a shell on App-B (10.20.1.10) through the bastion using ProxyJump.

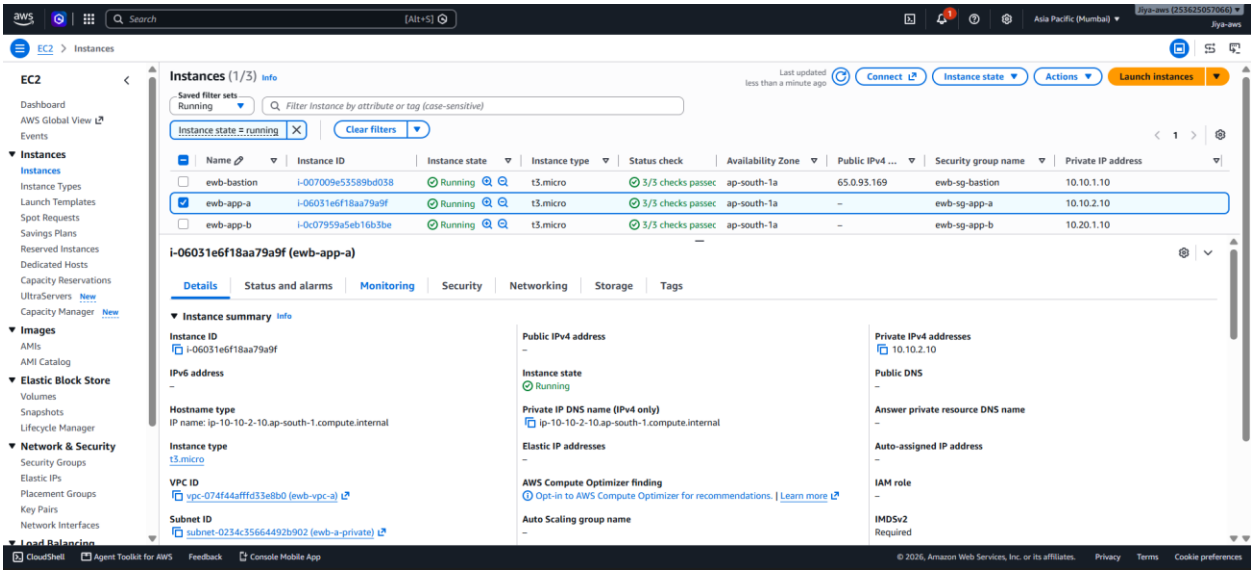


Figure 20. Test 05, App-A instance details: Public IPv4 address empty, private IP 10.10.2.10.

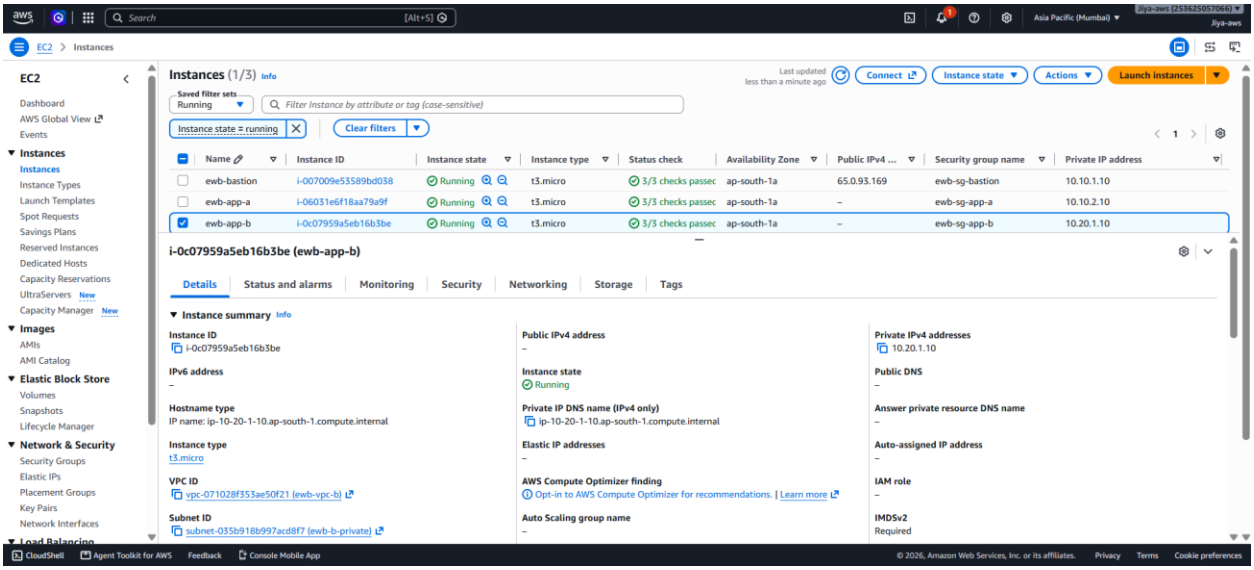


Figure 21. Test 05, App-B instance details: Public IPv4 address empty, private IP 10.20.1.10.

Controlled failure and recovery

Removing the ICMP Echo rule stopped the ping, and restoring it brought the ping back. I removed only the ICMP Echo request rule (source 10.10.2.10/32) from ewb-sg-app-b and kept the SSH rule. The ping from App-A to 10.20.1.10 then received no replies (100% packet loss). After I restored the exact same rule, the ping succeeded again with 4 packets received and 0% packet loss.

This shows that an Active peering connection and correct routes are not enough on their own. The security group on the destination must also allow the traffic.

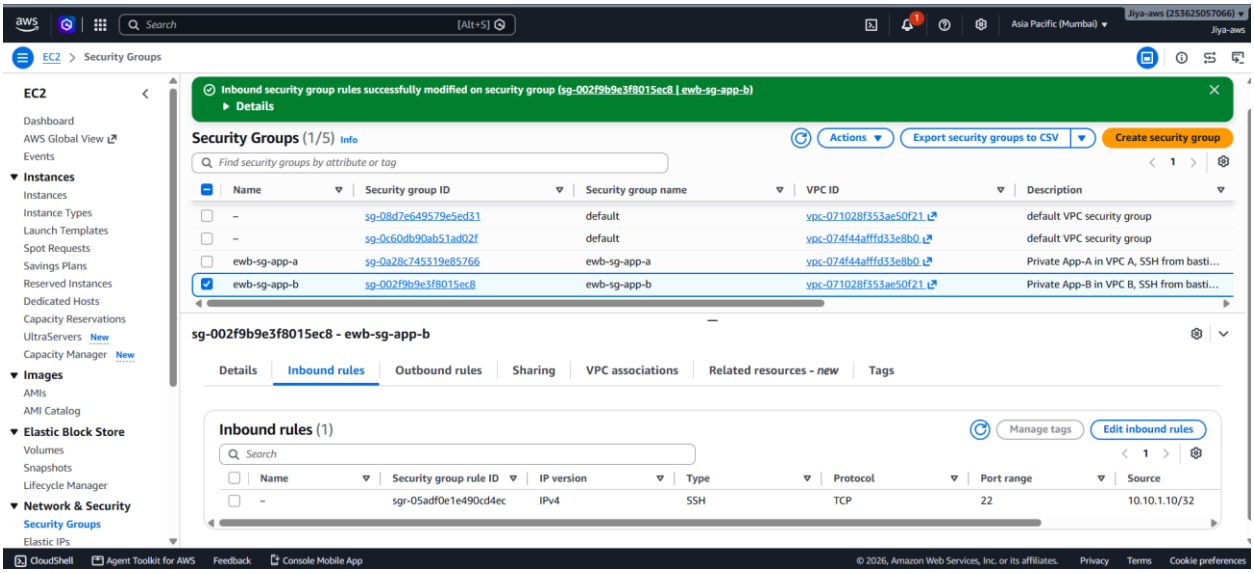


Figure 22. ewb-sg-app-b inbound rules with the ICMP Echo rule removed: only SSH, TCP 22 from 10.10.1.10/32 remains.

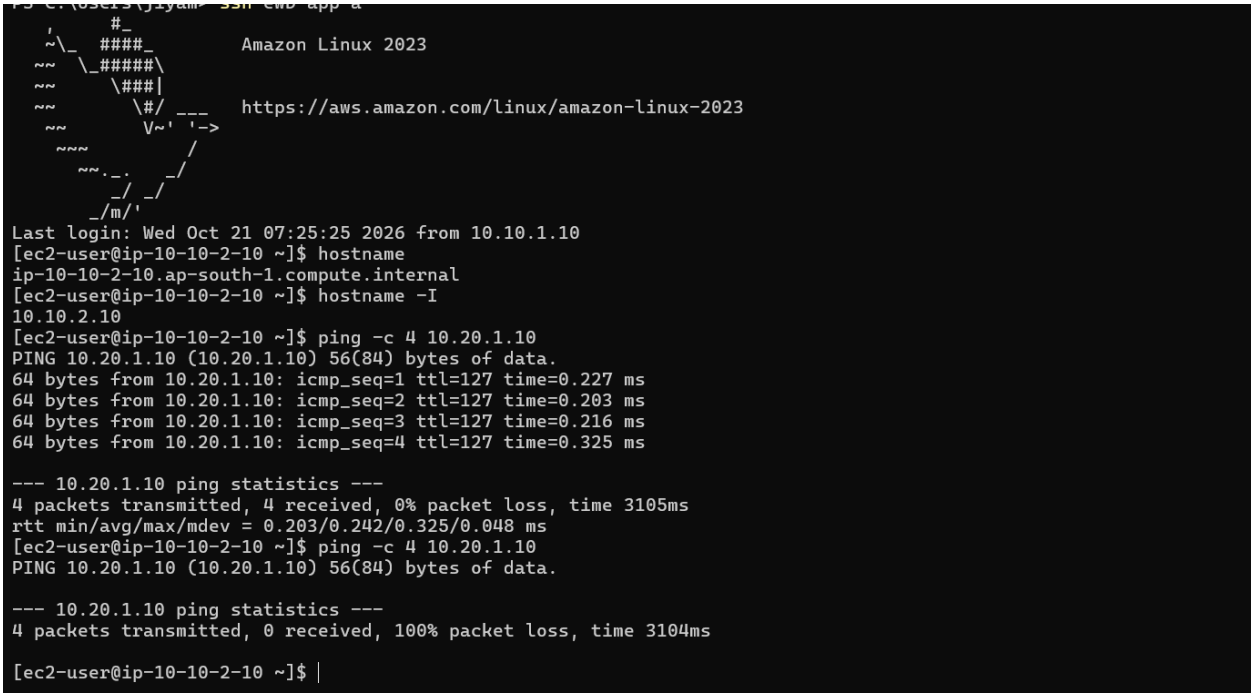


Figure 23. App-A: the earlier ping gets 4 of 4 replies; after the ICMP rule is removed, ping -c 4 10.20.1.10 gets 0 replies, 100% packet loss.

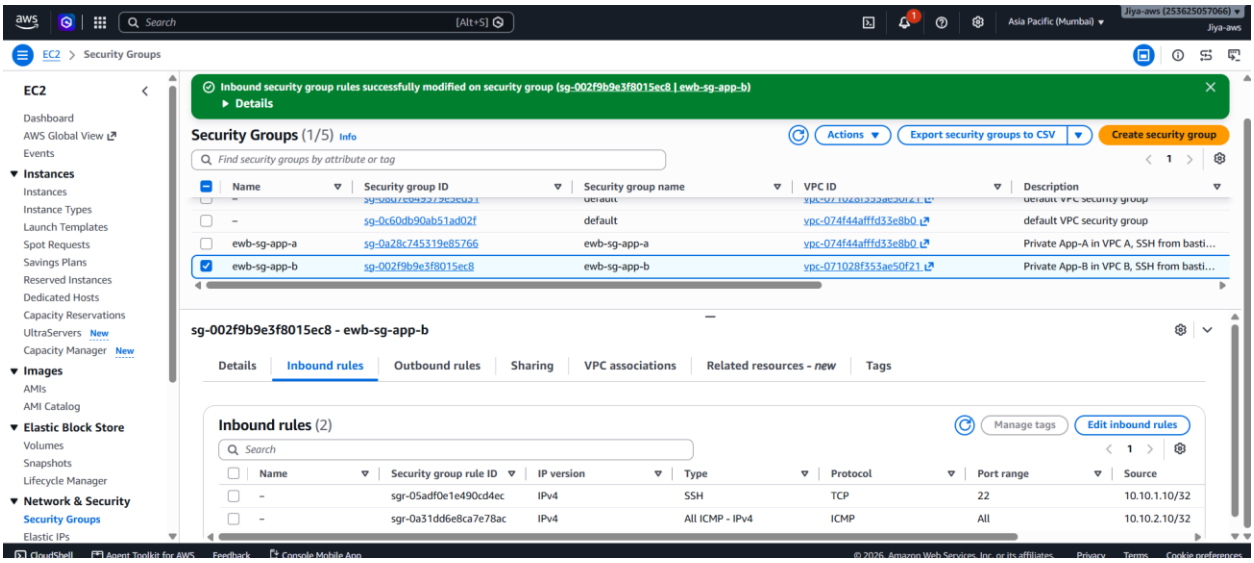


Figure 24. ewb-sg-app-b inbound rules with the ICMP rule restored for 10.10.2.10/32 alongside the SSH rule.

```
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.419 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.328 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.317 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.317 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3102ms
rtt min/avg/max/mdev = 0.317/0.345/0.419/0.042 ms
[ec2-user@ip-10-10-2-10 ~]$
```

Figure 25. App-A after recovery: ping -c 4 10.20.1.10 gets 4 of 4 replies, 0% packet loss.

Forward path and return path

A ping from App-A to App-B needs a route and a security rule in both directions.

- **Forward path:** App-A (10.10.2.10) sends a ping to 10.20.1.10. The route 10.20.0.0/16 in ewb-rt-a-private sends it to the peering connection, which delivers it to VPC B. ewb-sg-app-b allows the ICMP Echo request from 10.10.2.10/32.
- **Return path:** App-B replies to 10.10.2.10. The route 10.10.0.0/16 in ewb-rt-b-private sends the reply back through the peering connection. Security groups are stateful, so the reply is allowed automatically.
- **Bastion to App-B:** the bastion opens its own SSH connection to App-B, so ewb-rt-a-public also needs the 10.20.0.0/16 peering route, and ewb-sg-app-b allows SSH from the bastion's private IP (10.10.1.10/32).

Cleanup evidence

All resources created for this project were deleted in this order:

1. Instances (ewb-bastion, ewb-app-a, ewb-app-b)

2. Peering routes and the peering connection (ewb-a-b-peer)
3. Security groups
4. Subnets
5. Route tables
6. Internet gateway (ewb-igw-a)
7. VPCs (ewb-vpc-a and ewb-vpc-b)
8. Key pair (ewb-lab) and the local key file

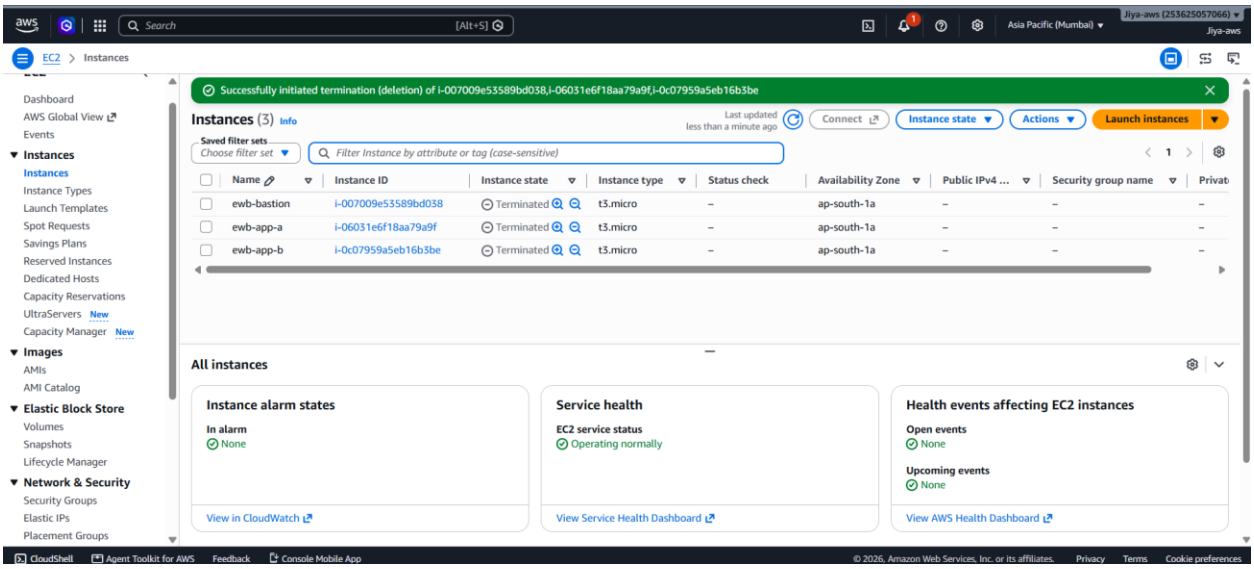


Figure 26. EC2 instances page: termination of ewb-bastion, ewb-app-a and ewb-app-b initiated, all three showing Terminated.

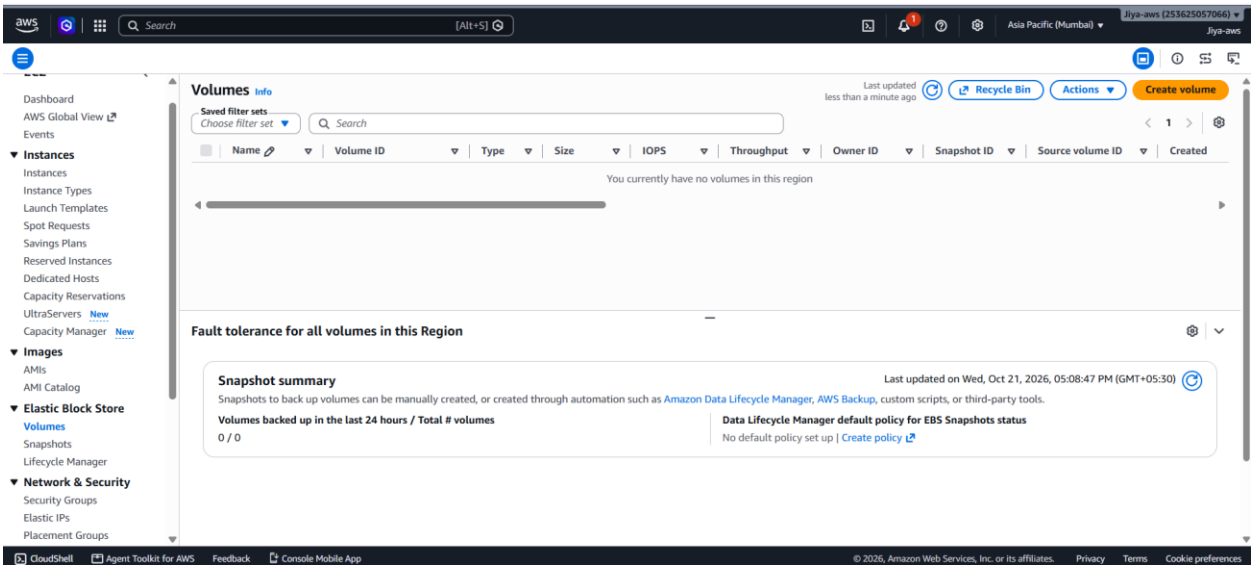


Figure 27. EC2 Volumes page: no volumes in the Region, so no lab volumes were left behind.

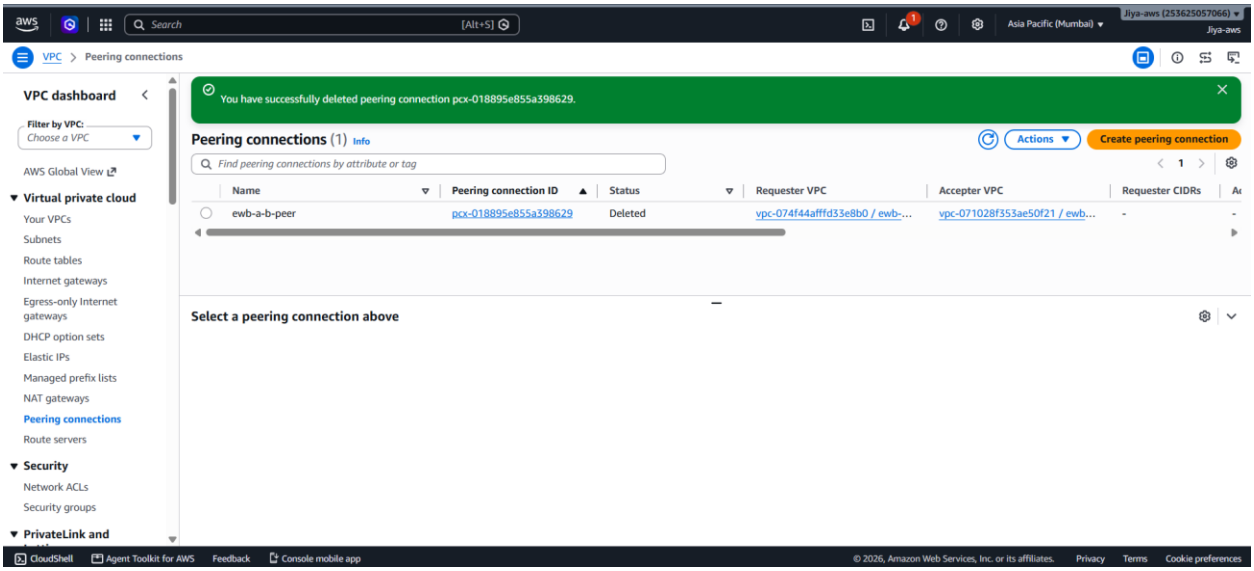


Figure 28. Peering connections page: pcx-018895e855a398629 (ewb-a-b-peer) deleted, status Deleted.

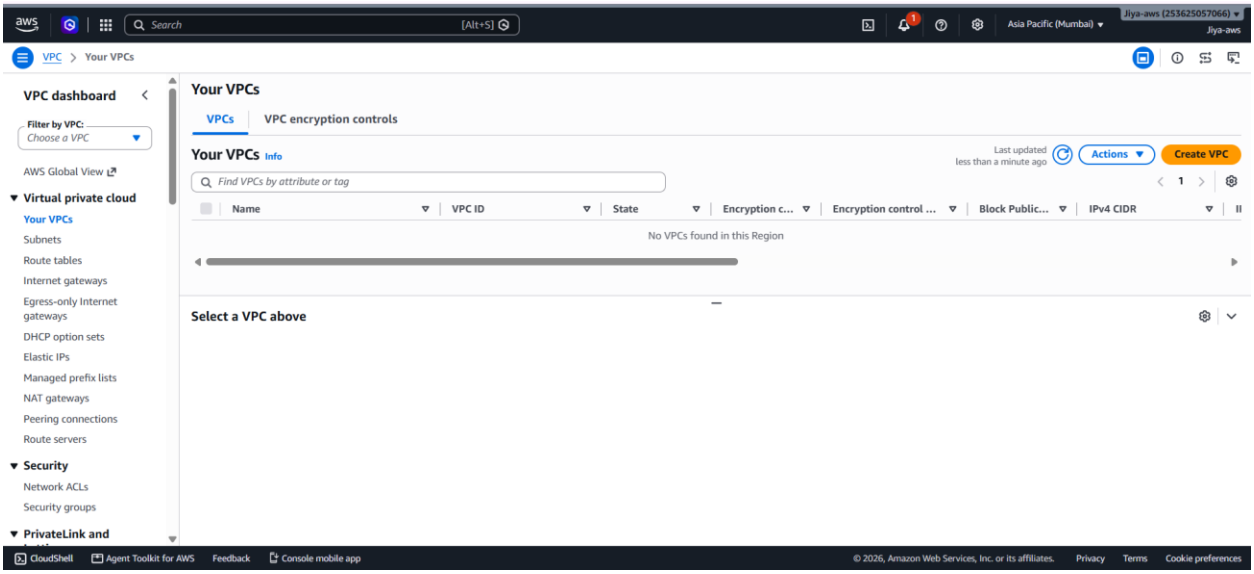


Figure 29. Your VPCs page: no VPCs found in the Region, so ewb-vpc-a and ewb-vpc-b are deleted.

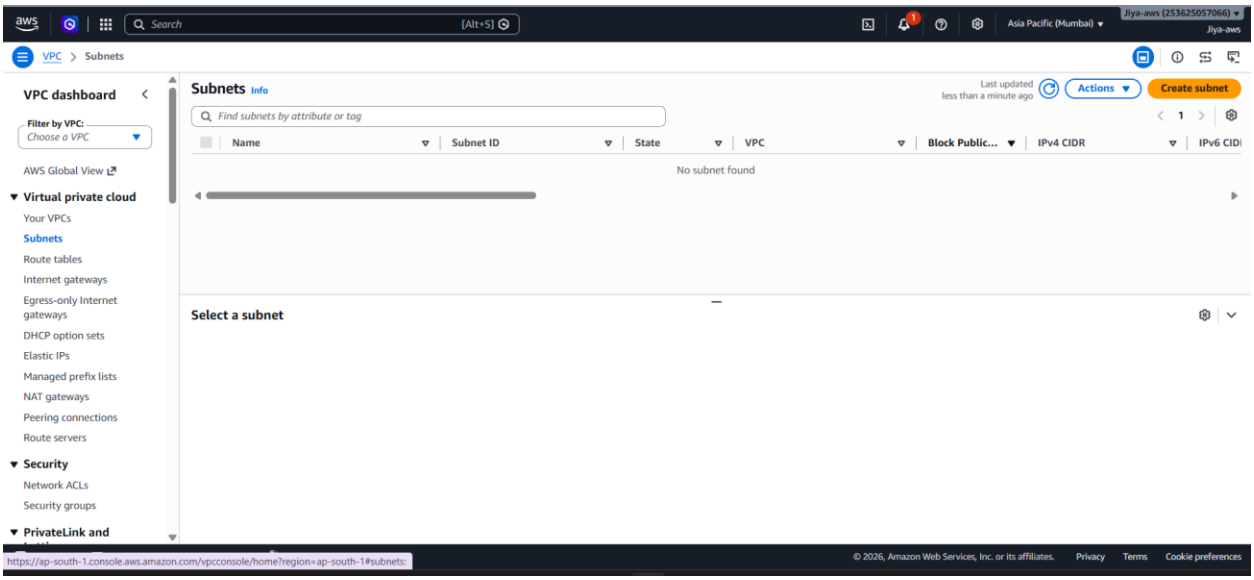


Figure 30. Subnets page: no subnets found.

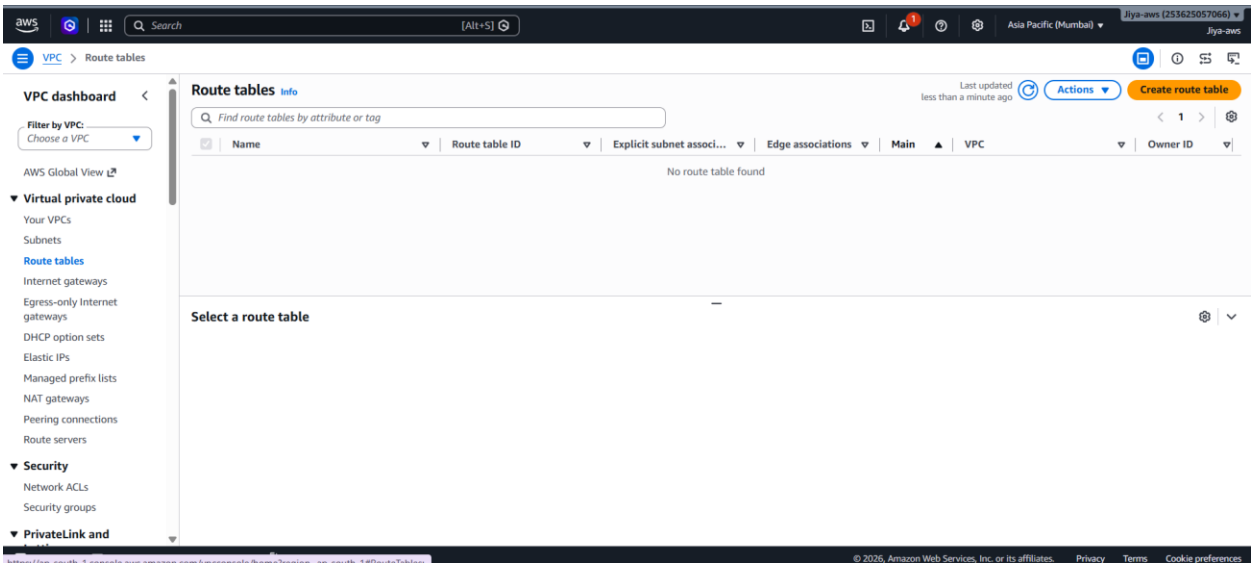


Figure 31. Route tables page: no route tables found.

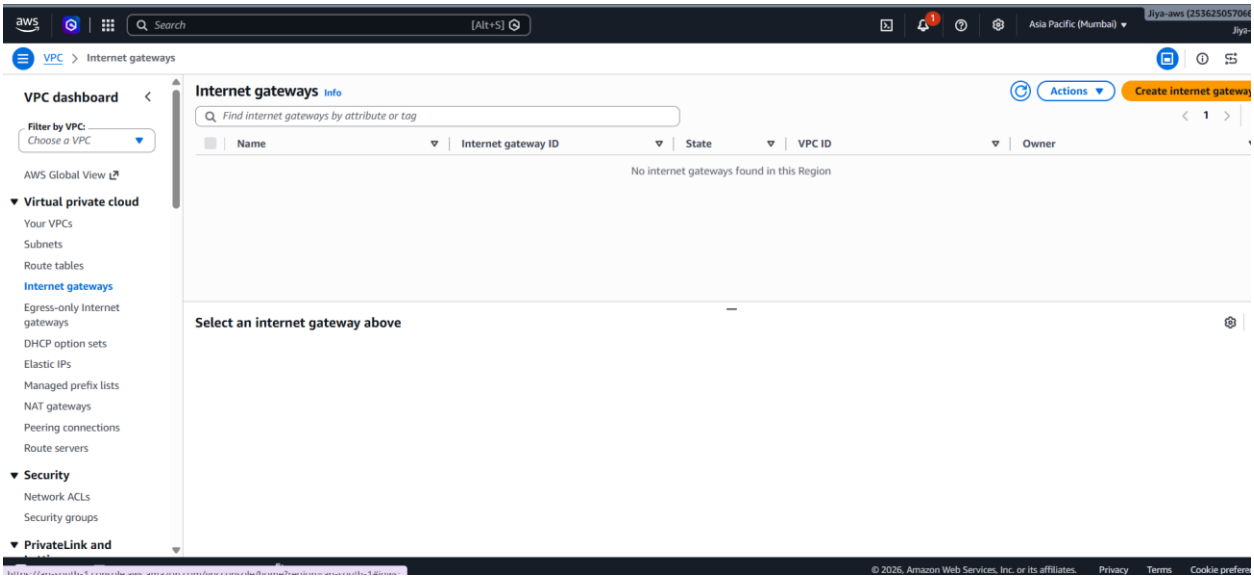


Figure 32. Internet gateways page: no internet gateways found, so ewb-igw-a is deleted.

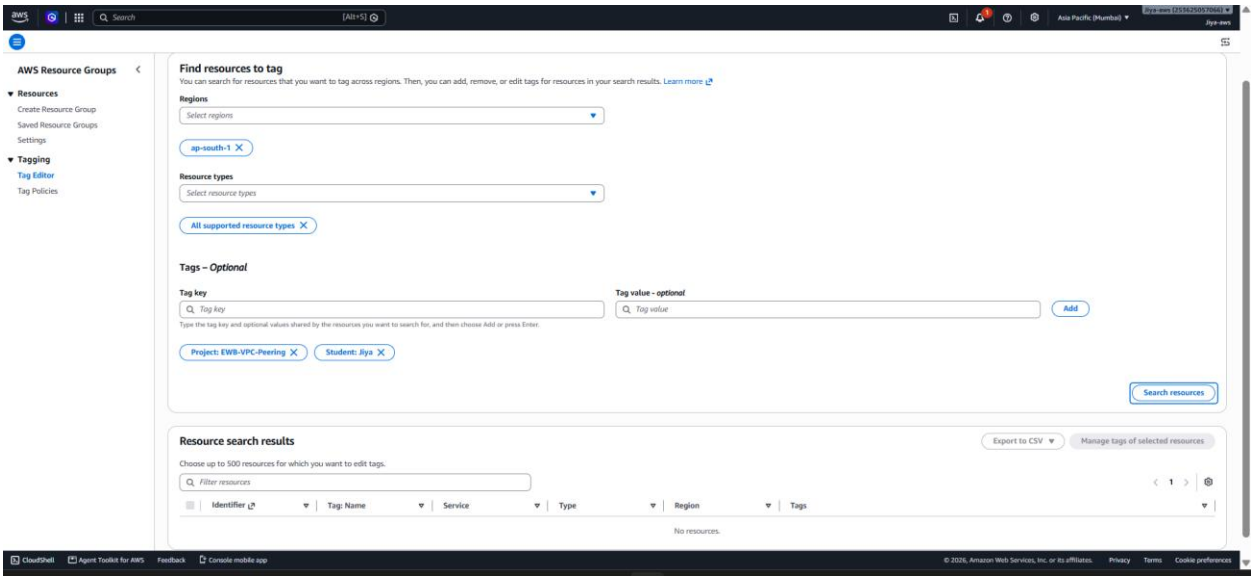


Figure 33. Tag Editor in ap-south-1, all supported resource types, tags Project = EWB-VPC-Peering and Student = Jiya: no resources found.

What I learned

VPC peering is only the link between two VPCs, and traffic uses it only when both sides have the right routes. I needed a route in each private route table (and in the public one for the bastion) pointing at the peering connection, because replies need a path back as well as the request.

Security groups still decide what is allowed, and the controlled failure showed this: removing the ICMP Echo rule stopped the ping even though the peering and routes were fine.

Using a bastion with ProxyJump let me reach both private servers without giving them public IPs or copying the key onto any instance. App-B sees the bastion's private IP as the SSH source, so that is the address its security group rule must allow.

I also learned that peering is non-transitive and gives VPC B no internet access through VPC A, and that SSH is strict about file permissions on keys and config files.